State	λ	0	1	x
0 initial	1 | λ | L	0 | 0 | R	0 | 1 | R	0 | x | R
1	31 | λ | S	2 | 0 | L	2 | 1 | L	31 | x | S
2	31 | λ | S	3 | 0 | D	4 | 1 | D	10 | x | R
3	6 | λ | L	6 | 0 | L	6 | 1 | L	6 | x | L
4	7 | λ | L	7 | 0 | L	7 | 1 | L	7 | x | L
5	31 | λ | S	8 | 0 | L	8 | 1 | L	31 | x | S
6	9 | 0 | U	31 | 0 | S	31 | 1 | S	31 | x | S
7	9 | 1 | U	31 | 0 | S	31 | 1 | S	31 | x | S
8	9 | x | U	31 | 0 | S	31 | 1 | S	31 | x | S
9	0 | λ | D	3 | 0 | D	4 | 1 | D	5 | x | D
10	31 | λ | S	11 | λ | L	12 | λ | L	31 | x | S
11	31 | λ | S	11 | λ | L	11 | λ | L	13 | λ | L
12	31 | λ | S	12 | λ | L	12 | λ | L	14 | λ | L
13	15 | 0 | R	13 | 0 | L	13 | 0 | L	31 | x | S
14	15 | 0 | R	14 | 0 | L	14 | 1 | L	31 | x | S
15	16 | λ | U	15 | 0 | R	15 | 1 | R	31 | x | S
16	18 | λ | D	17 | 0 | R	17 | 1 | R	31 | x | S
17	10 | λ | L	17 | 0 | R	17 | 1 | R	17 | x | R
18	19 | λ | L	31 | 0 | S	31 | 1 | S	31 | x | S
19	20 | λ | D	19 | 0 | L	19 | 1 | L	31 | x | S
20	30 | λ | U	21 | 0 | R	21 | 1 | R	31 | x | S
21	28 | λ | U	21 | 0 | R	21 | 1 | R	31 | x | S
22	19 | λ | D	24 | λ | D	25 | λ | D	31 | x | S
23	22 | λ | L	31 | 0 | S	31 | 1 | S	31 | x | S
24	23 | 0 | U	23 | 0 | U	23 | 1 | U	31 | x | S
25	23 | 1 | U	23 | 1 | U	27 | 0 | U	31 | x | S
26	25 | λ | D	25 | λ | D	29 | λ | D	31 | x | S
27	26 | λ | L	31 | 0 | S	31 | 1 | S	31 | x | S
28	22 | λ | L	28 | 0 | R	28 | 1 | R	31 | x | S
29	27 | 0 | U	27 | 0 | U	27 | 1 | U	31 | x | S
30	31 | λ | S	31 | 0 | S	31 | 1 | S	31 | x | S
31 halt	31 | λ | S	31 | 0 | S	31 | 1 | S	31 | x | S

Entry: next state | written symbol | motion

λ = blank - = unobserved S = stop